

MATH 501: REAL ANALYSIS

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1. AUGUST 25, 2026

Real analysis will be developed through measure theory, Lebesgue integration, and the study of function spaces such as L^p spaces.

1.1. Measures. The goal of a measure is to assign a length, area, or volume to subsets of $\mathbb{R}$, $\mathbb{R}^2$, or, more generally, $\mathbb{R}^n$. Ideally, we would like a function

$$m: 2^{\mathbb{R}^n} \longrightarrow [0, \infty]$$

with the following properties.

- (i) *Countable additivity:* if $E_1, E_2, \dots \subseteq \mathbb{R}^n$ are pairwise disjoint, then

$$m\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} m(E_j).$$

A weaker requirement is finite additivity, in which the identity is required only for finite pairwise disjoint families.

- (ii) *Invariance under rigid motions:* if

$$\tilde{E} = AE + b, \quad A \in O(n), \quad b \in \mathbb{R}^n,$$

then $m(\tilde{E}) = m(E)$. Here A represents a rotation or reflection, and b a translation.

- (iii) *Normalization:*

$$m([0, 1]^n) = 1.$$

Unfortunately, a measure with all these properties cannot be defined on every subset of $\mathbb{R}^n$.

Theorem 1 (Vitali). *Assuming the axiom of choice, there is no countably additive, rigid-motion-invariant map*

$$m: 2^{\mathbb{R}^n} \longrightarrow [0, \infty]$$

satisfying $m([0, 1]^n) = 1$.

Proof. It is enough to consider $n = 1$: if such a map existed on $\mathbb{R}^n$, then

$$\mu(E) = m(E \times [0, 1]^{n-1}), \quad E \subseteq \mathbb{R},$$

would have the required properties on $\mathbb{R}$. We therefore write m for this one-dimensional map.

First, $m(\{x\}) = 0$ for every $x \in \mathbb{R}$. Otherwise, translation invariance and countable additivity applied to a countably infinite set in $[0, 1]$ would contradict $m([0, 1]) = 1$. Consequently, $m([0, 1)) = 1$.

Define an equivalence relation on $[0, 1)$ by

$$x \sim y \iff x - y \in \mathbb{Q}.$$

Using the axiom of choice, select one representative from each equivalence class and let V be the set of representatives. Thus, for every $x \in [0, 1)$, there is a unique $y \in V$ such that $x - y \in \mathbb{Q}$.

For $t \in [0, 1)$, define the translate of V modulo 1 by

$$\begin{aligned} V_t &= V + t \pmod{1} \\ &= \{x + t : x \in V, x + t < 1\} \\ &\quad \cup \{x + t - 1 : x \in V, x + t \geq 1\}. \end{aligned}$$

Finite additivity and translation invariance give

$$m(V_t) = m(V).$$

The family indexed by $t \in [0, 1) \cap \mathbb{Q}$ is a partition of $[0, 1)$. Indeed, if $x \in [0, 1)$ and $y \in V$ represents the equivalence class of x , then either $t = x - y$ or $t = x - y + 1$ lies in $[0, 1) \cap \mathbb{Q}$ and $x \in V_t$. On the other hand, if $V_t \cap V_{t'} \neq \emptyset$, the uniqueness of representatives implies $t = t'$.

Countable additivity now yields

$$1 = m([0, 1)) = \sum_{t \in [0, 1) \cap \mathbb{Q}} m(V_t) = \sum_{t \in [0, 1) \cap \mathbb{Q}} m(V).$$

If $m(V) = 0$, the last sum is 0; if $m(V) > 0$, it is infinite. Either possibility is a contradiction. $\square$

One might try to rescue the construction by requiring only finite additivity together with invariance and normalization. In dimensions at least three, even this is impossible.

Theorem 2 (Banach–Tarski). *If $n \geq 3$, there is no finitely additive, rigid-motion-invariant map*

$$m: 2^{\mathbb{R}^n} \longrightarrow [0, \infty]$$

satisfying $m([0, 1]^n) = 1$.

The resolution is to give up the requirement that a measure be defined on *all* subsets of $\mathbb{R}^n$. In fact, measure theory can be developed on a general underlying set X once an appropriate collection of measurable subsets has been chosen.

1.2. Algebras and σ -algebras.

Definition 1 (Algebra). *An algebra on a set X is a nonempty collection $\mathcal{A} \subseteq 2^X$ such that*

- (a) *if $E_1, \dots, E_N \in \mathcal{A}$, then $\bigcup_{j=1}^N E_j \in \mathcal{A}$;*
- (b) *if $E \in \mathcal{A}$, then $E^c = X \setminus E \in \mathcal{A}$.*

Definition 2 (σ -algebra). *A σ -algebra on a set X is a nonempty collection $\mathcal{A} \subseteq 2^X$ such that*

- (a) *if $E_n \in \mathcal{A}$ for every $n \in \mathbb{N}$, then $\bigcup_{n=1}^{\infty} E_n \in \mathcal{A}$;*
- (b) *if $E \in \mathcal{A}$, then $E^c \in \mathcal{A}$.*

Because the collections in these definitions are nonempty and closed under complements, they contain both X and $\emptyset$. By De Morgan's laws, an algebra is closed under finite intersections and a σ -algebra is closed under countable intersections. For example,

$$\bigcap_{n=1}^{\infty} E_n = \left(\bigcup_{n=1}^{\infty} E_n^c \right)^c.$$

When an algebra is already known to be closed under finite set operations, it is enough to check closure under countable *disjoint* unions in order to prove that it is a σ -algebra. Given $E_n \in \mathcal{A}$, set

$$\tilde{E}_1 = E_1, \quad \tilde{E}_n = E_n \setminus \bigcup_{j=1}^{n-1} E_j \quad (n \geq 2).$$

Then the sets $\tilde{E}_n$ are pairwise disjoint and

$$\bigcup_{n=1}^{\infty} \tilde{E}_n = \bigcup_{n=1}^{\infty} E_n.$$

Example 1. *The collection $\{\emptyset, X\}$ is the smallest σ -algebra on X .*

Example 2. *The power set 2^X is the largest σ -algebra on X .*

Example 3 (Countable-cocountable σ -algebra). *The collection*

$$\mathcal{A} = \{E \subseteq X : E \text{ is countable or } E^c \text{ is countable}\}$$

is a σ -algebra on X .

A convenient way to specify a σ -algebra is to give a family of generators.

Definition 3 (Generated σ -algebra). *Let $\mathcal{E} \subseteq 2^X$. The σ -algebra generated by $\mathcal{E}$, denoted $\mathcal{M}(\mathcal{E})$, is the smallest σ -algebra containing $\mathcal{E}$.*

This object exists because 2^X is a σ -algebra containing $\mathcal{E}$, and an arbitrary intersection of σ -algebras is again a σ -algebra. Explicitly,

$$\mathcal{M}(\mathcal{E}) = \bigcap \{\mathcal{A} : \mathcal{A} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{E} \subseteq \mathcal{A}\}.$$

1.3. Borel σ -algebras.

Definition 4 (Borel σ -algebra). *Let X be a metric space, or more generally a topological space. The Borel σ -algebra $\mathcal{B}_X$ is the σ -algebra generated by the open subsets of X .*

For example, every open subset of $\mathbb{R}$ is a union of open intervals. The Borel σ -algebra contains every open and closed set, every G_δ set (a countable intersection of open sets), every F_σ set (a countable union of closed sets), and all sets obtained by continuing such countable operations. These classes form the Borel hierarchy.

Proposition 1. *The Borel σ -algebra $\mathcal{B}_{\mathbb{R}}$ is generated by any one of the following collections:*

- (a) $\{(a, b) : a < b\};$
- (b) $\{[a, b] : a < b\};$
- (c) $\{(-\infty, a] : a \in \mathbb{R}\}.$

Proof. Let the three collections above be $\mathcal{E}_1$, $\mathcal{E}_2$, and $\mathcal{E}_3$, respectively.

Every member of $\mathcal{E}_1$ is open, so $\mathcal{M}(\mathcal{E}_1) \subseteq \mathcal{B}_{\mathbb{R}}$. For the reverse inclusion, let $U \subseteq \mathbb{R}$ be open. For every $x \in U$, there is an $\varepsilon > 0$ such

that

$$(x - \varepsilon, x + \varepsilon) \subseteq U.$$

By the density of $\mathbb{Q}$ in $\mathbb{R}$, we can choose $p, q \in \mathbb{Q}$ such that

$$x - \varepsilon < p < x < q < x + \varepsilon.$$

Thus every $x \in U$ lies in an interval (p, q) with rational endpoints that is contained in U . It follows that

$$U = \bigcup \{(p, q) : p, q \in \mathbb{Q}, p < q, (p, q) \subseteq U\}.$$

The indexing collection is a subset of $\mathbb{Q}^2$, so it is countable. Therefore every open subset of $\mathbb{R}$ is a countable union of open intervals. In particular, every open set belongs to $\mathcal{M}(\mathcal{E}_1)$, proving

$$\mathcal{M}(\mathcal{E}_1) = \mathcal{B}_{\mathbb{R}}.$$

Every closed interval is Borel. In the other direction, for $a < b$,

$$(a, b) = \bigcup_{\substack{n \in \mathbb{N} \\ 2/n < b-a}} \left[a + \frac{1}{n}, b - \frac{1}{n} \right].$$

Thus $\mathcal{E}_1 \subseteq \mathcal{M}(\mathcal{E}_2)$, and consequently $\mathcal{M}(\mathcal{E}_2) = \mathcal{B}_{\mathbb{R}}$. Equivalently, the reverse containment can also be seen from

$$[a, b] = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b + \frac{1}{n} \right).$$

Finally, every left half-line in $\mathcal{E}_3$ is closed and hence Borel. Conversely,

$$(-\infty, b) = \bigcup_{n=1}^{\infty} \left(-\infty, b - \frac{1}{n} \right]$$

and therefore

$$(a, b) = (-\infty, b) \cap (-\infty, a]^c \in \mathcal{M}(\mathcal{E}_3).$$

It follows that $\mathcal{E}_1 \subseteq \mathcal{M}(\mathcal{E}_3)$, so $\mathcal{M}(\mathcal{E}_3) = \mathcal{B}_{\mathbb{R}}$ as well. $\square$

2. AUGUST 27, 2026

We continue our discussion of σ -algebras. Recall that a σ -algebra $\mathcal{M}$ on a set X contains $\emptyset$ and X , is closed under complements, and is closed under countable unions. If (X, d) is a metric space, then its Borel σ -algebra $\mathcal{B}_X$ is generated by the open subsets of X .

2.1. Product σ -algebras. Let I be an index set, let X_i be a nonempty set for each $i \in I$, and set

$$X = \prod_{i \in I} X_i.$$

For each $i \in I$, let $\mathcal{M}_i$ be a σ -algebra on X_i , and let

$$\pi_i: X \longrightarrow X_i, \quad \pi_i((x_j)_{j \in I}) = x_i,$$

be the coordinate projection.

Definition 5 (Product σ -algebra). *The product σ -algebra on X is*

$$\bigotimes_{i \in I} \mathcal{M}_i = \mathcal{M}(\{\pi_i^{-1}(E) : i \in I, E \in \mathcal{M}_i\}).$$

Thus it is the smallest σ -algebra that makes every coordinate projection measurable.

The generating sets $\pi_i^{-1}(E)$ are called *cylinder sets*. They restrict the i th coordinate to E and leave all other coordinates unrestricted.

Lemma 1. *If I is countable, then $\bigotimes_{i \in I} \mathcal{M}_i$ is generated by the measurable rectangles*

$$\left\{ \prod_{i \in I} E_i : E_i \in \mathcal{M}_i \right\}.$$

Proof. Since I is countable, every measurable rectangle satisfies

$$\prod_{i \in I} E_i = \bigcap_{i \in I} \pi_i^{-1}(E_i) \in \bigotimes_{i \in I} \mathcal{M}_i.$$

Conversely, each cylinder set is a measurable rectangle. Indeed,

$$\pi_j^{-1}(E_j) = \prod_{i \in I} F_i, \quad F_i = \begin{cases} E_j, & i = j, \\ X_i, & i \neq j. \end{cases}$$

The two collections therefore generate the same σ -algebra. $\square$

The generators of the coordinate σ -algebras can be used directly to generate the product.

Proposition 2. *Suppose*

$$\mathcal{M}_i = \mathcal{M}(\mathcal{G}_i), \quad \mathcal{G}_i \subseteq 2^{X_i},$$

for every $i \in I$.

(a) *The product σ -algebra is generated by*

$$\mathcal{F} = \{\pi_i^{-1}(G) : i \in I, G \in \mathcal{G}_i\}.$$

(b) *If I is countable and $\mathcal{G}_i^* = \mathcal{G}_i \cup \{X_i\}$, then the product σ -algebra is generated by*

$$\left\{ \prod_{i \in I} G_i : G_i \in \mathcal{G}_i^* \right\}.$$

Proof. For part (a), every member of $\mathcal{F}$ belongs to $\bigotimes_{i \in I} \mathcal{M}_i$, so

$$\mathcal{M}(\mathcal{F}) \subseteq \bigotimes_{i \in I} \mathcal{M}_i.$$

For the reverse inclusion, fix $i \in I$ and consider

$$\mathcal{A}_i = \{E \subseteq X_i : \pi_i^{-1}(E) \in \mathcal{M}(\mathcal{F})\}.$$

Preimages preserve complements and countable unions, so $\mathcal{A}_i$ is a σ -algebra. It contains $\mathcal{G}_i$ and hence contains $\mathcal{M}(\mathcal{G}_i) = \mathcal{M}_i$. Therefore every cylinder $\pi_i^{-1}(E)$ with $E \in \mathcal{M}_i$ belongs to $\mathcal{M}(\mathcal{F})$, which proves the reverse inclusion.

For part (b), each cylinder from part (a) is a rectangle whose other coordinates are X_j . Conversely, since I is countable, every rectangle in the displayed collection is the countable intersection of the sets $\pi_i^{-1}(G_i)$. When $G_i = X_i$ this set is all of X ; otherwise it is a cylinder from part (a). The conclusion follows. $\square$

2.2. Products of Borel σ -algebras. Let (X_i, d_i) , $i = 1, \dots, n$, be metric spaces. On

$$X = \prod_{i=1}^n X_i$$

define the maximum metric

$$d(x, y) = \max_{1 \leq i \leq n} d_i(x_i, y_i).$$

Proposition 3. *With the notation above,*

$$\bigotimes_{i=1}^n \mathcal{B}_{X_i} \subseteq \mathcal{B}_X.$$

If every (X_i, d_i) is separable, then equality holds:

$$\bigotimes_{i=1}^n \mathcal{B}_{X_i} = \mathcal{B}_X.$$

Proof. Each coordinate projection $\pi_i: X \rightarrow X_i$ is continuous. If $U_i \subseteq X_i$ is open, then $\pi_i^{-1}(U_i)$ is open in X . Since $\mathcal{B}_{X_i}$ is generated by the open subsets of X_i , Proposition 2 gives

$$\bigotimes_{i=1}^n \mathcal{B}_{X_i} \subseteq \mathcal{B}_X.$$

Now suppose that each X_i is separable, and choose a countable dense subset $D_i \subseteq X_i$. The collection

$$\mathcal{G}_i = \{B_{d_i}(p, q) : p \in D_i, q \in \mathbb{Q}, q > 0\}$$

is a countable basis for X_i and hence generates $\mathcal{B}_{X_i}$. The finite product $D = \prod_{i=1}^n D_i$ is countable and dense in X . Moreover,

$$B_d(x, q) = \prod_{i=1}^n B_{d_i}(x_i, q).$$

Thus every ball in X whose center lies in D and whose radius is a positive rational belongs to $\bigotimes_{i=1}^n \mathcal{B}_{X_i}$. These balls form a countable basis for X , so every open subset of X is a countable union of such balls. Hence

$$\mathcal{B}_X \subseteq \bigotimes_{i=1}^n \mathcal{B}_{X_i},$$

which completes the proof. $\square$

Corollary 1. *For every $n \in \mathbb{N}$,*

$$\mathcal{B}_{\mathbb{R}^n} = \bigotimes_{i=1}^n \mathcal{B}_{\mathbb{R}}.$$

For example, $\mathcal{B}_{\mathbb{R}^n}$ is generated by each of the following collections:

- open rectangles $\prod_{i=1}^n (a_i, b_i)$;
- closed rectangles $\prod_{i=1}^n [a_i, b_i]$; and
- lower orthants $\prod_{i=1}^n (-\infty, a_i]$.

2.3. Measures.

Definition 6 (Measure). *Let $\mathcal{M}$ be a σ -algebra on a set X . A function*

$$\mu: \mathcal{M} \longrightarrow [0, \infty]$$

is a measure if

- (i) $\mu(\emptyset) = 0$;
- (ii) *whenever $E_1, E_2, \dots \in \mathcal{M}$ are pairwise disjoint,*

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i).$$

The second property is called countable additivity. If it is required only for finite pairwise disjoint families, then μ is called finitely additive.

The sets in $\mathcal{M}$ are called *measurable sets*, and the triple $(X, \mathcal{M}, \mu)$ is called a *measure space*. A measure μ is

- *finite* if $\mu(X) < \infty$;
- *σ -finite* if $X = \bigcup_{i=1}^{\infty} E_i$ for some $E_i \in \mathcal{M}$ with $\mu(E_i) < \infty$ for every i .

A set $N \in \mathcal{M}$ with $\mu(N) = 0$ is called a *null set*. A statement is said to hold *almost everywhere*, abbreviated a.e., if it holds outside some null set.

2.4. Examples of measures.

Example 4 (Weighted counting measure). *Let X be nonempty, let $\mathcal{M} = 2^X$, and let $f: X \rightarrow [0, \infty]$. Define*

$$\mu(E) = \sum_{x \in E} f(x) := \sup \left\{ \sum_{x \in F} f(x) : F \subseteq E, F \text{ finite} \right\}.$$

Then μ is a measure on 2^X .

If $f(x) = 1$ for every $x \in X$, then μ is the counting measure,

$$\mu(E) = \#E.$$

If $x_0 \in X$ and

$$f(x) = \begin{cases} 1, & x = x_0, \\ 0, & x \neq x_0, \end{cases}$$

then

$$\mu(E) = \begin{cases} 1, & x_0 \in E, \\ 0, & x_0 \notin E. \end{cases}$$

This is the point mass, or Dirac measure, at x_0 .

Example 5 (Countable-cocountable measure). *Let X be uncountable, and let $\mathcal{M}$ be the countable-cocountable σ -algebra. Then*

$$\mu(E) = \begin{cases} 0, & E \text{ is countable,} \\ 1, & E^c \text{ is countable} \end{cases}$$

defines a measure on $(X, \mathcal{M})$.

Example 6 (Finite additivity is not enough). *Let $X = \mathbb{N}$ and $\mathcal{M} = 2^{\mathbb{N}}$. Define*

$$\nu(E) = \begin{cases} 0, & E \text{ is finite,} \\ \infty, & E \text{ is infinite.} \end{cases}$$

Then ν is finitely additive. It is not a measure, because

$$\nu\left(\bigcup_{n=1}^{\infty} \{n\}\right) = \infty \quad \text{but} \quad \sum_{n=1}^{\infty} \nu(\{n\}) = 0.$$

2.5. Basic properties of measures.

Theorem 3. *Let $(X, \mathcal{M}, \mu)$ be a measure space.*

(a) Monotonicity: if $E, F \in \mathcal{M}$ and $E \subseteq F$, then

$$\mu(E) \leq \mu(F).$$

(b) Countable subadditivity: if $E_1, E_2, \dots \in \mathcal{M}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) \leq \sum_{i=1}^{\infty} \mu(E_i).$$

(c) Continuity from below: if

$$E_1 \subseteq E_2 \subseteq \dots, \quad E_i \in \mathcal{M},$$

then

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \lim_{i \rightarrow \infty} \mu(E_i).$$

(d) Continuity from above: if

$$E_1 \supseteq E_2 \supseteq \dots, \quad E_i \in \mathcal{M}, \quad \mu(E_1) < \infty,$$

then

$$\mu\left(\bigcap_{i=1}^{\infty} E_i\right) = \lim_{i \rightarrow \infty} \mu(E_i).$$

Proof. For part (a), write F as the disjoint union

$$F = E \cup (F \setminus E).$$

Countable additivity gives

$$\mu(F) = \mu(E) + \mu(F \setminus E) \geq \mu(E).$$

For part (b), define

$$F_1 = E_1, \quad F_k = E_k \setminus \bigcup_{j=1}^{k-1} E_j \quad (k \geq 2).$$

The sets F_k are pairwise disjoint, $F_k \subseteq E_k$, and

$$\bigcup_{k=1}^{\infty} F_k = \bigcup_{k=1}^{\infty} E_k.$$

Therefore countable additivity and part (a) imply

$$\mu \left(\bigcup_{k=1}^{\infty} E_k \right) = \sum_{k=1}^{\infty} \mu(F_k) \leq \sum_{k=1}^{\infty} \mu(E_k).$$

For part (c), set

$$F_1 = E_1, \quad F_k = E_k \setminus E_{k-1} \quad (k \geq 2).$$

The sets F_k are pairwise disjoint, and for each N ,

$$E_N = \bigcup_{k=1}^N F_k, \quad \bigcup_{i=1}^{\infty} E_i = \bigcup_{k=1}^{\infty} F_k.$$

It follows that

$$\begin{aligned} \mu \left(\bigcup_{i=1}^{\infty} E_i \right) &= \sum_{k=1}^{\infty} \mu(F_k) \\ &= \lim_{N \rightarrow \infty} \sum_{k=1}^N \mu(F_k) \\ &= \lim_{N \rightarrow \infty} \mu(E_N). \end{aligned}$$

For part (d), define

$$F_i = E_1 \setminus E_i.$$

Since E_i decreases, F_i increases, and

$$\bigcup_{i=1}^{\infty} F_i = E_1 \setminus \bigcap_{i=1}^{\infty} E_i.$$

By part (c),

$$\mu \left(E_1 \setminus \bigcap_{i=1}^{\infty} E_i \right) = \lim_{i \rightarrow \infty} \mu(E_1 \setminus E_i).$$

Because $E_i \subseteq E_1$ and $\mu(E_1) < \infty$, finite additivity gives

$$\mu(E_1 \setminus E_i) = \mu(E_1) - \mu(E_i)$$

and

$$\mu \left(E_1 \setminus \bigcap_{i=1}^{\infty} E_i \right) = \mu(E_1) - \mu \left(\bigcap_{i=1}^{\infty} E_i \right).$$

Substituting these identities into the preceding limit and cancelling the finite quantity $\mu(E_1)$ yields

$$\mu\left(\bigcap_{i=1}^{\infty} E_i\right) = \lim_{i \rightarrow \infty} \mu(E_i).$$

□

Remark 1 (Necessity of the finite-measure hypothesis). The assumption $\mu(E_1) < \infty$ in Theorem 3(d) cannot be omitted. Let

$$X = \mathbb{N}, \quad \mathcal{M} = 2^{\mathbb{N}},$$

let μ be counting measure, and define

$$E_i = \{i, i+1, i+2, \dots\}.$$

Then

$$E_1 \supseteq E_2 \supseteq E_3 \supseteq \dots, \quad \bigcap_{i=1}^{\infty} E_i = \emptyset.$$

Indeed, for any $n \in \mathbb{N}$, we have $n \notin E_i$ whenever $i > n$. Consequently,

$$\mu\left(\bigcap_{i=1}^{\infty} E_i\right) = \mu(\emptyset) = 0.$$

On the other hand, every E_i is infinite, so

$$\mu(E_i) = \infty \quad \text{for every } i, \quad \lim_{i \rightarrow \infty} \mu(E_i) = \infty.$$

Thus

$$\mu\left(\bigcap_{i=1}^{\infty} E_i\right) = 0 \neq \infty = \lim_{i \rightarrow \infty} \mu(E_i).$$

This does not contradict part (d), because $\mu(E_1) = \mu(\mathbb{N}) = \infty$.

3. SEPTEMBER 1, 2026

Recall that a measure on a σ -algebra $\mathcal{M}$ over a set X is a map

$$\mu: \mathcal{M} \longrightarrow [0, \infty]$$

such that $\mu(\emptyset) = 0$ and

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

whenever $A_1, A_2, \dots \in \mathcal{M}$ are pairwise disjoint. For Lebesgue measure m on $\mathbb{R}$, the normalization is $m([0, 1]) = 1$.

3.1. Complete measure spaces.

Definition 7 (Complete measure). *A measure μ on a σ -algebra $\mathcal{M}$ over X is complete if every subset of a null set is measurable. That is, if $E \subseteq N$ for some $N \in \mathcal{M}$ with $\mu(N) = 0$, then $E \in \mathcal{M}$.*

By monotonicity, every such set E also satisfies

$$0 \leq \mu(E) \leq \mu(N) = 0,$$

so $\mu(E) = 0$.

Every measure space has a canonical completion obtained by adjoining all subsets of its null sets.

Proposition 4 (Completion of a measure). *Let $(X, \mathcal{M}, \mu)$ be a measure space, and let*

$$\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}.$$

Define

$$\overline{\mathcal{M}} = \{E \cup F : E \in \mathcal{M}, F \subseteq N \text{ for some } N \in \mathcal{N}\}.$$

Then the following statements hold.

- (i) *The collection $\overline{\mathcal{M}}$ is a σ -algebra on X containing $\mathcal{M}$.*
- (ii) *There is a unique measure $\overline{\mu}$ on $\overline{\mathcal{M}}$ extending μ . It is given by*

$$\overline{\mu}(E \cup F) = \mu(E),$$

where $E \in \mathcal{M}$ and F is contained in a member of $\mathcal{N}$. Moreover, $\overline{\mu}$ is complete.

Proof. The collections $\mathcal{M}$ and $\mathcal{N}$ are closed under countable unions. For $\mathcal{N}$, this follows from countable subadditivity: if $N_i \in \mathcal{N}$, then

$$0 \leq \mu \left(\bigcup_{i=1}^{\infty} N_i \right) \leq \sum_{i=1}^{\infty} \mu(N_i) = 0.$$

Hence $\overline{\mathcal{M}}$ is closed under countable unions. Indeed, if $E_i \in \mathcal{M}$ and $F_i \subseteq N_i \in \mathcal{N}$, then

$$\bigcup_{i=1}^{\infty} (E_i \cup F_i) = \left(\bigcup_{i=1}^{\infty} E_i \right) \cup \left(\bigcup_{i=1}^{\infty} F_i \right),$$

and $\bigcup_i F_i \subseteq \bigcup_i N_i \in \mathcal{N}$.

To check closure under complements, let $E \in \mathcal{M}$ and $F \subseteq N \in \mathcal{N}$. Replacing N by $N \setminus E$ and F by $F \setminus E$ does not change $E \cup F$, so we may assume that $E \cap N = \emptyset$. In that case,

$$(E \cup F)^c = (E \cup N)^c \cup (N \setminus F).$$

The first set on the right belongs to $\mathcal{M}$, and the second is contained in the null set N . Thus $(E \cup F)^c$ belongs to $\overline{\mathcal{M}}$. Since $\emptyset \in \overline{\mathcal{M}}$, this proves that $\overline{\mathcal{M}}$ is a σ -algebra. Taking $F = \emptyset$ shows that $\mathcal{M} \subseteq \overline{\mathcal{M}}$.

We next show that the formula for $\bar{\mu}$ is well defined. Suppose

$$E_1 \cup F_1 = E_2 \cup F_2, \quad E_i \in \mathcal{M}, \quad F_i \subseteq N_i \in \mathcal{N}.$$

Then $E_1 \subseteq E_2 \cup N_2$, and therefore

$$\mu(E_1) \leq \mu(E_2 \cup N_2) \leq \mu(E_2) + \mu(N_2) = \mu(E_2).$$

By symmetry, $\mu(E_2) \leq \mu(E_1)$, so $\mu(E_1) = \mu(E_2)$.

Clearly $\bar{\mu}(\emptyset) = 0$. Suppose that $A_i = E_i \cup F_i$ are pairwise disjoint members of $\overline{\mathcal{M}}$, where $E_i \in \mathcal{M}$ and $F_i \subseteq N_i \in \mathcal{N}$. The sets E_i are pairwise disjoint, so

$$\begin{aligned} \bar{\mu} \left(\bigcup_{i=1}^{\infty} A_i \right) &= \mu \left(\bigcup_{i=1}^{\infty} E_i \right) \\ &= \sum_{i=1}^{\infty} \mu(E_i) \end{aligned}$$

$$= \sum_{i=1}^{\infty} \bar{\mu}(A_i).$$

Thus $\bar{\mu}$ is a measure, and it extends μ .

If $A = E \cup F \in \overline{\mathcal{M}}$ and $\bar{\mu}(A) = 0$, then $\mu(E) = 0$. Since $A \subseteq E \cup N$ and $E \cup N$ is a measurable null set, every subset of A belongs to $\overline{\mathcal{M}}$. Hence $\bar{\mu}$ is complete.

Finally, let ν be any measure on $\overline{\mathcal{M}}$ that extends μ . If $A = E \cup F$, with $E \in \mathcal{M}$ and $F \subseteq N \in \mathcal{N}$, then

$$E \subseteq A \subseteq E \cup N.$$

Monotonicity and the fact that ν extends μ give

$$\mu(E) = \nu(E) \leq \nu(A) \leq \nu(E \cup N) = \mu(E \cup N) = \mu(E).$$

Thus $\nu(A) = \mu(E) = \bar{\mu}(A)$, which proves uniqueness. $\square$

The measure $\bar{\mu}$ is called the *completion* of μ .

3.2. Outer measures. The goal is to construct measures from simple geometric pieces. For example, to estimate the area of a set $E \subseteq \mathbb{R}^2$, one can cover E by squares of side length s and consider

$$\inf_{s>0} (\text{number of squares in the cover}) s^2.$$

More generally, one covers a set by countably many simple pieces and takes the infimum of the sums of their assigned sizes.

A related guiding principle is that the “reasonable” subsets $E \subseteq [0, 1]^2$ should be those for which outer and inner area agree. If the unit square has area 1, its inner estimate for E is

$$1 - m^*([0, 1]^2 \setminus E).$$

Definition 8 (Outer measure). *An outer measure on a set X is a map*

$$\mu^*: 2^X \longrightarrow [0, \infty]$$

satisfying the following properties.

- (i) $\mu^*(\emptyset) = 0$.
- (ii) If $A \subseteq B \subseteq X$, then $\mu^*(A) \leq \mu^*(B)$.

(iii) For every sequence $A_1, A_2, \dots \subseteq X$,

$$\mu^* \left(\bigcup_{j=1}^{\infty} A_j \right) \leq \sum_{j=1}^{\infty} \mu^*(A_j).$$

Unlike a measure, an outer measure is defined on every subset of X ; its countable subadditivity therefore applies without a measurability assumption.

Proposition 5 (Outer measure from coverings). *Let $\mathcal{E} \subseteq 2^X$ contain $\emptyset$ and X , and let*

$$\rho: \mathcal{E} \longrightarrow [0, \infty]$$

satisfy $\rho(\emptyset) = 0$. For $A \subseteq X$, define

$$\mu^*(A) = \inf \left\{ \sum_{j=1}^{\infty} \rho(E_j) : E_j \in \mathcal{E}, \quad A \subseteq \bigcup_{j=1}^{\infty} E_j \right\}.$$

Then μ^ is an outer measure on X .*

Remark 2. Even when $E \in \mathcal{E}$, it need not be true that $\mu^*(E) = \rho(E)$. A more efficient countable cover may have smaller total ρ -size than the single-set cover by E .

Example 7. *Let $X = \mathbb{R}$ and take*

$$\mathcal{E} = \{\emptyset, \mathbb{R}\} \cup \{[a, b] : a, b \in \mathbb{R}, a \leq b\}.$$

Define

$$\rho(\emptyset) = 0, \quad \rho(\mathbb{R}) = \infty, \quad \rho([a, b]) = (b - a)^2.$$

Although $\rho([0, 1]) = 1$, the induced outer measure satisfies $\mu^([0, 1]) = 0$. Indeed, for every $n \in \mathbb{N}$,*

$$[0, 1] = \bigcup_{j=0}^{n-1} \left[\frac{j}{n}, \frac{j+1}{n} \right],$$

and the total ρ -size of this cover is

$$\sum_{j=0}^{n-1} \left(\frac{1}{n} \right)^2 = \frac{1}{n}.$$

Hence $0 \leq \mu^([0, 1]) \leq 1/n$ for every n , which gives the claim.*

Proof of the proposition. The defining collection of sums is nonempty because any set $A \subseteq X$ is covered by the sequence $X, \emptyset, \emptyset, \dots$. Thus $\mu^*(A)$ is well defined as an element of $[0, \infty]$.

The sequence $\emptyset, \emptyset, \dots$ covers $\emptyset$, so

$$0 \leq \mu^*(\emptyset) \leq \sum_{j=1}^{\infty} \rho(\emptyset) = 0.$$

If $A \subseteq B$, then every countable cover of B by members of $\mathcal{E}$ is also a cover of A . Taking infima over the respective collections of covers gives $\mu^*(A) \leq \mu^*(B)$.

It remains to prove countable subadditivity. Let $A_j \subseteq X$ for $j \in \mathbb{N}$. If $\sum_j \mu^*(A_j) = \infty$, there is nothing to prove. Otherwise, fix $\varepsilon > 0$. For every j , choose $E_{j,k} \in \mathcal{E}$, $k \in \mathbb{N}$, such that

$$A_j \subseteq \bigcup_{k=1}^{\infty} E_{j,k} \quad \text{and} \quad \sum_{k=1}^{\infty} \rho(E_{j,k}) \leq \mu^*(A_j) + \varepsilon 2^{-j}.$$

The doubly indexed collection $\{E_{j,k} : j, k \in \mathbb{N}\}$ is countable and covers $\bigcup_j A_j$. Consequently,

$$\begin{aligned} \mu^*\left(\bigcup_{j=1}^{\infty} A_j\right) &\leq \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \rho(E_{j,k}) \\ &\leq \sum_{j=1}^{\infty} (\mu^*(A_j) + \varepsilon 2^{-j}) \\ &= \sum_{j=1}^{\infty} \mu^*(A_j) + \varepsilon. \end{aligned}$$

Letting $\varepsilon \downarrow 0$ proves the result. $\square$

Definition 9 (Measurability with respect to an outer measure). *Let μ^* be an outer measure on X . A set $A \subseteq X$ is μ^* -measurable if, for every $E \subseteq X$,*

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

Countable subadditivity always gives

$$\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c),$$

so the content of the measurability criterion is the reverse inequality. When $\mu^*(E) < \infty$, the quantity

$$\mu^*(E) - \mu^*(E \cap A^c)$$

can be viewed as the inner measure of $E \cap A$ relative to E .

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Recall that if μ^* is an outer measure on X , a set $A \subseteq X$ is μ^* -measurable if

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

for every $E \subseteq X$. By countable subadditivity, it is enough to prove the inequality

$$\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

4.1. The Carathéodory construction.

Theorem 4 (Carathéodory). *Let μ^* be an outer measure on X , and set*

$$\mathcal{M}^* = \{A \subseteq X : A \text{ is } \mu^*\text{-measurable}\}.$$

Then $\mathcal{M}^$ is a σ -algebra, and*

$$\mu = \mu^*|_{\mathcal{M}^*}$$

is a complete measure on $\mathcal{M}^$.*

Proof. We first show that $\mathcal{M}^*$ is an algebra. The empty set is measurable because

$$\mu^*(E) = \mu^*(E \cap \emptyset) + \mu^*(E \cap X)$$

for every $E \subseteq X$. The defining condition is symmetric in A and A^c , so

$$A \in \mathcal{M}^* \implies A^c \in \mathcal{M}^*.$$

Now let $A, B \in \mathcal{M}^*$ and fix $E \subseteq X$. Applying the measurability criterion first to A and then to B gives

$$\begin{aligned}\mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\ &= \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) \\ &\quad + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c).\end{aligned}$$

The first three sets on the last two lines have union $E \cap (A \cup B)$. Therefore countable subadditivity implies

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c).$$

Thus $A \cup B \in \mathcal{M}^*$, and $\mathcal{M}^*$ is an algebra.

This argument also establishes finite additivity on disjoint measurable sets. Indeed, if $A, B \in \mathcal{M}^*$ are disjoint, apply the measurability of A with $E = A \cup B$ to obtain

$$\mu^*(A \cup B) = \mu^*((A \cup B) \cap A) + \mu^*((A \cup B) \cap A^c) = \mu^*(A) + \mu^*(B).$$

To prove closure under countable unions, first let $A_1, A_2, \dots \in \mathcal{M}^*$ be pairwise disjoint, and set

$$B = \bigcup_{i=1}^{\infty} A_i, \quad B_n = \bigcup_{i=1}^n A_i.$$

Since $\mathcal{M}^*$ is an algebra, $B_n \in \mathcal{M}^*$. Repeatedly applying the measurability criterion and using the disjointness of the A_i gives, for every $E \subseteq X$,

$$\mu^*(E) = \sum_{i=1}^n \mu^*(E \cap A_i) + \mu^*(E \cap B_n^c).$$

Because $B^c \subseteq B_n^c$, monotonicity yields

$$\mu^*(E) \geq \sum_{i=1}^n \mu^*(E \cap A_i) + \mu^*(E \cap B^c).$$

Letting $n \rightarrow \infty$ and then using countable subadditivity, we find

$$\mu^*(E) \geq \sum_{i=1}^{\infty} \mu^*(E \cap A_i) + \mu^*(E \cap B^c)$$

$$\begin{aligned}
&\geq \mu^* \left(\bigcup_{i=1}^{\infty} (E \cap A_i) \right) + \mu^*(E \cap B^c) \\
&= \mu^*(E \cap B) + \mu^*(E \cap B^c).
\end{aligned}$$

Hence $B \in \mathcal{M}^*$. An arbitrary countable union can be disjointified by replacing A_i with

$$A_i \setminus \bigcup_{j=1}^{i-1} A_j.$$

It follows that $\mathcal{M}^*$ is a σ -algebra.

The same computation proves countable additivity. If the measurable sets A_i are pairwise disjoint and $B = \bigcup_i A_i$, then countable subadditivity gives

$$\mu^*(B) \leq \sum_{i=1}^{\infty} \mu^*(A_i).$$

On the other hand, for every n ,

$$\mu^*(B) = \sum_{i=1}^n \mu^*(A_i) + \mu^*(B \setminus B_n) \geq \sum_{i=1}^n \mu^*(A_i).$$

Letting $n \rightarrow \infty$ proves

$$\mu^*(B) = \sum_{i=1}^{\infty} \mu^*(A_i).$$

Thus $\mu = \mu^*|_{\mathcal{M}^*}$ is a measure.

Finally, suppose that $A \subseteq X$ and $\mu^*(A) = 0$. For every $E \subseteq X$, subadditivity and monotonicity give

$$\mu^*(E \cap A) + \mu^*(E \cap A^c) \leq 0 + \mu^*(E) = \mu^*(E).$$

Hence $A \in \mathcal{M}^*$. In particular, if $N \in \mathcal{M}^*$ satisfies $\mu(N) = 0$, then every subset $A \subseteq N$ has outer measure zero and therefore belongs to $\mathcal{M}^*$. This proves that μ is complete. $\square$

4.2. Premeasures. The covering construction from the preceding lecture begins with a set function ρ defined on a family of elementary sets.

The example $\rho([a, b]) = (b - a)^2$ showed that the resulting outer measure may fail to agree with ρ , even on elementary sets. To prevent this collapse, we require the initial set function to behave like a measure.

Definition 10 (Premeasure). *Let $\mathcal{A}$ be an algebra on X . A map*

$$\mu_0: \mathcal{A} \longrightarrow [0, \infty]$$

is a premeasure if

- (i) $\mu_0(\emptyset) = 0$;
- (ii) if $A_1, A_2, \dots \in \mathcal{A}$ are pairwise disjoint and $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$, then

$$\mu_0\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu_0(A_i).$$

Thus a premeasure has the same additivity property as a measure, but its domain is only required to be an algebra, rather than a σ -algebra.

Example 8 (Half-open intervals). *Let $X = \mathbb{R}$. Consider intervals of the forms*

$$(a, b], \quad (-\infty, b], \quad (a, \infty),$$

together with $\emptyset$, where $a, b \in \mathbb{R}$ and $a < b$. The collection $\mathcal{A}$ of finite unions of such intervals is an algebra. Moreover,

$$\mathcal{M}(\mathcal{A}) = \mathcal{B}_{\mathbb{R}}.$$

This algebra will later be used to construct Lebesgue measure.

The next proposition identifies the hypotheses under which the outer measure construction preserves the initial set function.

Proposition 6 (Extension of a premeasure). *Let μ_0 be a premeasure on an algebra $\mathcal{A}$ over X . For every $E \subseteq X$, define*

$$\mu^*(E) = \inf \left\{ \sum_{i=1}^{\infty} \mu_0(A_i) : A_i \in \mathcal{A}, \quad E \subseteq \bigcup_{i=1}^{\infty} A_i \right\}.$$

Then the following statements hold.

- (i) $\mu^*|_{\mathcal{A}} = \mu_0$.

(ii) Every member of $\mathcal{A}$ is μ^* -measurable.

Consequently, $\mathcal{M}(\mathcal{A}) \subseteq \mathcal{M}^*$, and the measure $\mu^*|_{\mathcal{M}(\mathcal{A})}$ extends μ_0 .

Proof. Let $A \in \mathcal{A}$. The sequence $A, \emptyset, \emptyset, \dots$ is a cover of A , so

$$\mu^*(A) \leq \mu_0(A).$$

Conversely, suppose that $A_i \in \mathcal{A}$ and $A \subseteq \bigcup_{i=1}^{\infty} A_i$. Define

$$B_1 = A \cap A_1, \quad B_k = A \cap \left(A_k \setminus \bigcup_{i=1}^{k-1} A_i \right) \quad (k \geq 2).$$

Then the sets B_k are pairwise disjoint members of $\mathcal{A}$, and

$$A = \bigcup_{k=1}^{\infty} B_k.$$

Since $B_k \subseteq A_k$, monotonicity of the premeasure gives

$$\mu_0(A) = \sum_{k=1}^{\infty} \mu_0(B_k) \leq \sum_{k=1}^{\infty} \mu_0(A_k).$$

Taking the infimum over all such covers yields $\mu_0(A) \leq \mu^*(A)$. Therefore $\mu^*(A) = \mu_0(A)$, proving part (i).

For part (ii), fix $A \in \mathcal{A}$ and $E \subseteq X$. If $\mu^*(E) = \infty$, the desired measurability identity follows at once from countable subadditivity. Suppose instead that $\mu^*(E) < \infty$, and let $\varepsilon > 0$. Choose $A_1, A_2, \dots \in \mathcal{A}$ such that

$$E \subseteq \bigcup_{i=1}^{\infty} A_i \quad \text{and} \quad \mu^*(E) + \varepsilon \geq \sum_{i=1}^{\infty} \mu_0(A_i).$$

Since μ_0 is finitely additive,

$$\begin{aligned} \sum_{i=1}^{\infty} \mu_0(A_i) &= \sum_{i=1}^{\infty} (\mu_0(A_i \cap A) + \mu_0(A_i \cap A^c)) \\ &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c). \end{aligned}$$

Hence

$$\mu^*(E) + \varepsilon \geq \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

Letting $\varepsilon \downarrow 0$ proves that A is μ^* -measurable.

By Theorem 4, $\mathcal{M}^*$ is a σ -algebra. Since it contains $\mathcal{A}$, it also contains $\mathcal{M}(\mathcal{A})$. Part (i) then shows that the restriction of μ^* to $\mathcal{M}(\mathcal{A})$ extends μ_0 . $\square$

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We continue the construction of measures from premeasures. Recall that if μ_0 is a premeasure on an algebra $\mathcal{A}$, then

$$\mu^*(E) = \inf \left\{ \sum_{j=1}^{\infty} \mu_0(A_j) : A_j \in \mathcal{A}, \quad E \subseteq \bigcup_{j=1}^{\infty} A_j \right\}$$

defines an outer measure. Every member of $\mathcal{A}$ is μ^* -measurable, and

$$\mu = \mu^*|_{\mathcal{M}(\mathcal{A})}$$

is a measure extending μ_0 .

5.1. Uniqueness of the extension. The extension is unique on sets of finite measure. Under a σ -finiteness assumption, it is unique everywhere.

Theorem 5 (Uniqueness of the premeasure extension). *Let μ_0 be a premeasure on an algebra $\mathcal{A}$, and let μ be the measure on $\mathcal{M}(\mathcal{A})$ obtained from the induced outer measure. Suppose that ν is another measure on $\mathcal{M}(\mathcal{A})$ extending μ_0 .*

- (a) *If $E \in \mathcal{M}(\mathcal{A})$ and $\mu(E) < \infty$, then $\nu(E) = \mu(E)$.*
- (b) *If μ_0 is σ -finite, then $\nu = \mu$ on $\mathcal{M}(\mathcal{A})$.*

Proof. We divide the proof of part (a) into three steps.

First, let $E \in \mathcal{M}(\mathcal{A})$ and suppose that $E \subseteq \bigcup_{j=1}^{\infty} A_j$, where $A_j \in \mathcal{A}$. Since ν extends μ_0 , countable subadditivity gives

$$\nu(E) \leq \nu \left(\bigcup_{j=1}^{\infty} A_j \right) \leq \sum_{j=1}^{\infty} \nu(A_j) = \sum_{j=1}^{\infty} \mu_0(A_j).$$

Taking the infimum over all such covers yields

$$(1) \quad \nu(E) \leq \mu^*(E) = \mu(E).$$

Second, suppose that

$$A = \bigcup_{j=1}^{\infty} A_j, \quad A_j \in \mathcal{A}.$$

Set $B_n = \bigcup_{j=1}^n A_j$. Then $B_n \in \mathcal{A}$ and $B_n \uparrow A$. Continuity from below, together with the fact that μ and ν both extend μ_0 , gives

$$\nu(A) = \lim_{n \rightarrow \infty} \nu(B_n) = \lim_{n \rightarrow \infty} \mu(B_n) = \mu(A).$$

Third, let $E \in \mathcal{M}(\mathcal{A})$ satisfy $\mu(E) < \infty$, and fix $\varepsilon > 0$. By the definition of μ^* , there are $A_j \in \mathcal{A}$ such that, with $A = \bigcup_{j=1}^{\infty} A_j$,

$$E \subseteq A, \quad \mu(A) \leq \sum_{j=1}^{\infty} \mu_0(A_j) < \mu(E) + \varepsilon.$$

In particular, $\mu(A \setminus E) < \varepsilon$. By the second step and (1),

$$\begin{aligned} \mu(E) &\leq \mu(A) = \nu(A) \\ &= \nu(E) + \nu(A \setminus E) \\ &\leq \nu(E) + \mu(A \setminus E) < \nu(E) + \varepsilon. \end{aligned}$$

Letting $\varepsilon \downarrow 0$ shows that $\mu(E) \leq \nu(E)$. Together with (1), this proves part (a).

For part (b), σ -finiteness provides sets $A_j \in \mathcal{A}$ such that

$$X = \bigcup_{j=1}^{\infty} A_j, \quad \mu_0(A_j) < \infty.$$

By replacing A_j with $A_j \setminus \bigcup_{i=1}^{j-1} A_i$, we may assume that the sets are pairwise disjoint. If $E \in \mathcal{M}(\mathcal{A})$, then $\mu(E \cap A_j) \leq \mu_0(A_j) < \infty$, so part (a) applies to each $E \cap A_j$. Therefore

$$\mu(E) = \sum_{j=1}^{\infty} \mu(E \cap A_j) = \sum_{j=1}^{\infty} \nu(E \cap A_j) = \nu(E).$$

□

5.2. Lebesgue–Stieltjes measures on $\mathbb{R}$. Let μ be a finite Borel measure on $\mathbb{R}$. Its *distribution function* is

$$(2) \quad F(x) = \mu((-\infty, x]), \quad x \in \mathbb{R}.$$

This function records the measure of every left half-line.

Proposition 7. *The distribution function F has the following properties.*

- (i) *It is increasing.*
- (ii) *It is right-continuous.*
- (iii) *For every $a < b$,*

$$\mu((a, b]) = F(b) - F(a).$$

Proof. If $x \leq y$, then $(-\infty, x] \subseteq (-\infty, y]$, so monotonicity of μ gives $F(x) \leq F(y)$.

If $x_n \downarrow x$, then

$$(-\infty, x_n] \downarrow (-\infty, x].$$

Since μ is finite, continuity from above gives

$$\lim_{n \rightarrow \infty} F(x_n) = \lim_{n \rightarrow \infty} \mu((-\infty, x_n]) = \mu((-\infty, x]) = F(x).$$

Hence F is right-continuous. Finally,

$$(-\infty, b] = (-\infty, a] \dot{\cup} (a, b],$$

and finite additivity gives

$$\mu((a, b]) = \mu((-\infty, b]) - \mu((-\infty, a]) = F(b) - F(a).$$

□

Example 9 (Distribution function of a Dirac measure). *Let $\mu = \delta_{x_0}$ be the Dirac measure at $x_0 \in \mathbb{R}$. Then*

$$F(x) = \delta_{x_0}((-\infty, x]) = \begin{cases} 0, & x < x_0, \\ 1, & x \geq x_0. \end{cases}$$

Thus an atom of a measure appears as a jump in its distribution function.

Remark 3 (Measures finite on bounded sets). A Borel measure need not be finite on all of $\mathbb{R}$. If μ is finite on every bounded Borel set, define the normalized distribution function

$$\tilde{F}(x) = \begin{cases} \mu((0, x]), & x > 0, \\ 0, & x = 0, \\ -\mu((x, 0]), & x < 0. \end{cases}$$

Then $\tilde{F}$ is increasing and right-continuous, and

$$\mu((a, b]) = \tilde{F}(b) - \tilde{F}(a)$$

for all $a < b$. Adding a constant to $\tilde{F}$ does not change these interval increments.

We now reverse the preceding construction at the level of the algebra of half-open intervals. Let $F: \mathbb{R} \rightarrow \mathbb{R}$ be increasing and right-continuous, and set

$$F(-\infty) = \lim_{x \rightarrow -\infty} F(x), \quad F(\infty) = \lim_{x \rightarrow \infty} F(x),$$

where these limits are allowed to be infinite. Let $\mathcal{A}$ be the algebra of finite unions of intervals of the forms

$$(a, b], \quad (-\infty, b], \quad (a, \infty),$$

together with $\emptyset$. Equivalently, its members are finite unions of intervals $(a, b] \cap \mathbb{R}$ with $-\infty \leq a < b \leq \infty$.

Proposition 8 (The Lebesgue–Stieltjes premeasure). *Suppose that*

$$E = \bigcup_{j=1}^N (a_j, b_j]$$

is a representation of $E \in \mathcal{A}$ as a finite union of pairwise disjoint half-open intervals, with extended endpoints allowed as above. Define

$$(3) \quad \mu_0(E) = \sum_{j=1}^N (F(b_j) - F(a_j)), \quad \mu_0(\emptyset) = 0.$$

Then $\mu_0: \mathcal{A} \rightarrow [0, \infty]$ is well defined and is a premeasure.

Proof. We first show that the value in (3) is independent of the chosen representation. Suppose that a half-open interval is partitioned into finitely many pairwise disjoint half-open intervals:

$$(a, b] = \bigcup_{j=1}^N (a_j, b_j].$$

After relabeling, their endpoints satisfy

$$a = a_1 < b_1 = a_2 < b_2 = \cdots < b_{N-1} = a_N < b_N = b.$$

Hence the sum telescopes:

$$\sum_{j=1}^N (F(b_j) - F(a_j)) = F(b) - F(a).$$

The same argument applies when an endpoint is infinite, using the conventions for $F(-\infty)$ and $F(\infty)$.

Now suppose that

$$\bigcup_{i=1}^N I_i = \bigcup_{j=1}^M J_j,$$

where both families consist of pairwise disjoint half-open intervals. The nonempty intersections $I_i \cap J_j$ partition both sides. By the preceding telescoping identity,

$$\begin{aligned} \sum_{i=1}^N \mu_0(I_i) &= \sum_{i=1}^N \sum_{j=1}^M \mu_0(I_i \cap J_j) \\ &= \sum_{j=1}^M \sum_{i=1}^N \mu_0(I_i \cap J_j) = \sum_{j=1}^M \mu_0(J_j). \end{aligned}$$

Thus μ_0 is well defined; the same calculation also gives finite additivity.

It remains to prove countable additivity. After decomposing all sets involved into their interval components and intersecting with each component of the union, it is enough to consider pairwise disjoint half-open intervals I_j such that

$$(a, b] = \bigcup_{j=1}^{\infty} I_j.$$

Finite additivity and monotonicity give, for every n ,

$$\mu_0((a, b]) \geq \mu_0\left(\bigcup_{j=1}^n I_j\right) = \sum_{j=1}^n \mu_0(I_j).$$

Letting $n \rightarrow \infty$ yields

$$(4) \quad \mu_0((a, b]) \geq \sum_{j=1}^{\infty} \mu_0(I_j).$$

For the reverse inequality, first suppose that $a, b \in \mathbb{R}$ and write $I_j = (a_j, b_j]$. Fix $\varepsilon > 0$. Right-continuity provides $0 < \delta < b - a$ and $\delta_j > 0$ such that

$$F(a + \delta) - F(a) < \varepsilon, \quad F(b_j + \delta_j) - F(b_j) < \varepsilon 2^{-j}.$$

The open intervals $(a_j, b_j + \delta_j)$ cover the compact interval $[a + \delta, b]$, so finitely many of them suffice. After discarding redundant intervals, relabel this finite subcover from left to right with indices $1, \dots, r$. Its total enlargement still satisfies

$$\sum_{j=1}^r (F(b_j + \delta_j) - F(b_j)) < \varepsilon,$$

and the ordering may be chosen so that

$$a_1 < a + \delta, \quad a_{j+1} < b_j + \delta_j \quad (1 \leq j < r), \quad b < b_r + \delta_r.$$

Monotonicity of F then gives

$$\begin{aligned} F(b) - F(a) &< F(b) - F(a + \delta) + \varepsilon \\ &\leq F(b_r + \delta_r) - F(a_1) + \varepsilon \\ &= F(b_r + \delta_r) - F(a_r) + \sum_{j=1}^{r-1} (F(a_{j+1}) - F(a_j)) + \varepsilon \\ &\leq \sum_{j=1}^r (F(b_j + \delta_j) - F(a_j)) + \varepsilon \\ &< \sum_{j=1}^r (F(b_j) - F(a_j)) + 2\varepsilon \end{aligned}$$

$$\leq \sum_{j=1}^{\infty} \mu_0(I_j) + 2\varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves the reverse of (4) for bounded intervals.

If the interval on the left is unbounded, apply the bounded case to its truncations. For example,

$$(-m, b] \uparrow (-\infty, b], \quad \mu_0((-m, b]) = F(b) - F(-m) \uparrow F(b) - F(-\infty),$$

while the bounded case gives

$$\mu_0((-m, b]) = \sum_{j=1}^{\infty} \mu_0(I_j \cap (-m, b]) \leq \sum_{j=1}^{\infty} \mu_0(I_j).$$

Letting $m \rightarrow \infty$ proves the desired inequality for $(-\infty, b]$. The same argument uses $(a, m] \uparrow (a, \infty)$ or $(-m, m] \uparrow \mathbb{R}$ for the other unbounded cases. Therefore

$$\mu_0\left(\bigcup_{j=1}^{\infty} I_j\right) = \sum_{j=1}^{\infty} \mu_0(I_j),$$

whenever the union belongs to $\mathcal{A}$, so μ_0 is a premeasure. $\square$

The premeasure extension theorem from the preceding lecture therefore applies to μ_0 on $\mathcal{A}$.

Remark 4 (Two Lebesgue–Stieltjes examples). The Borel measure determined by the increments $F(b) - F(a)$ is denoted by μ_F , and its completion is called the *Lebesgue–Stieltjes measure* associated with F .

(i) For $F(x) = x$,

$$\mu_F((a, b]) = b - a.$$

The completed measure is Lebesgue measure on $\mathbb{R}$.

(ii) For the right-continuous step function

$$F(x) = \mathbf{1}_{[x_0, \infty)}(x),$$

the associated measure is the Dirac measure $\mu_F = \delta_{x_0}$.